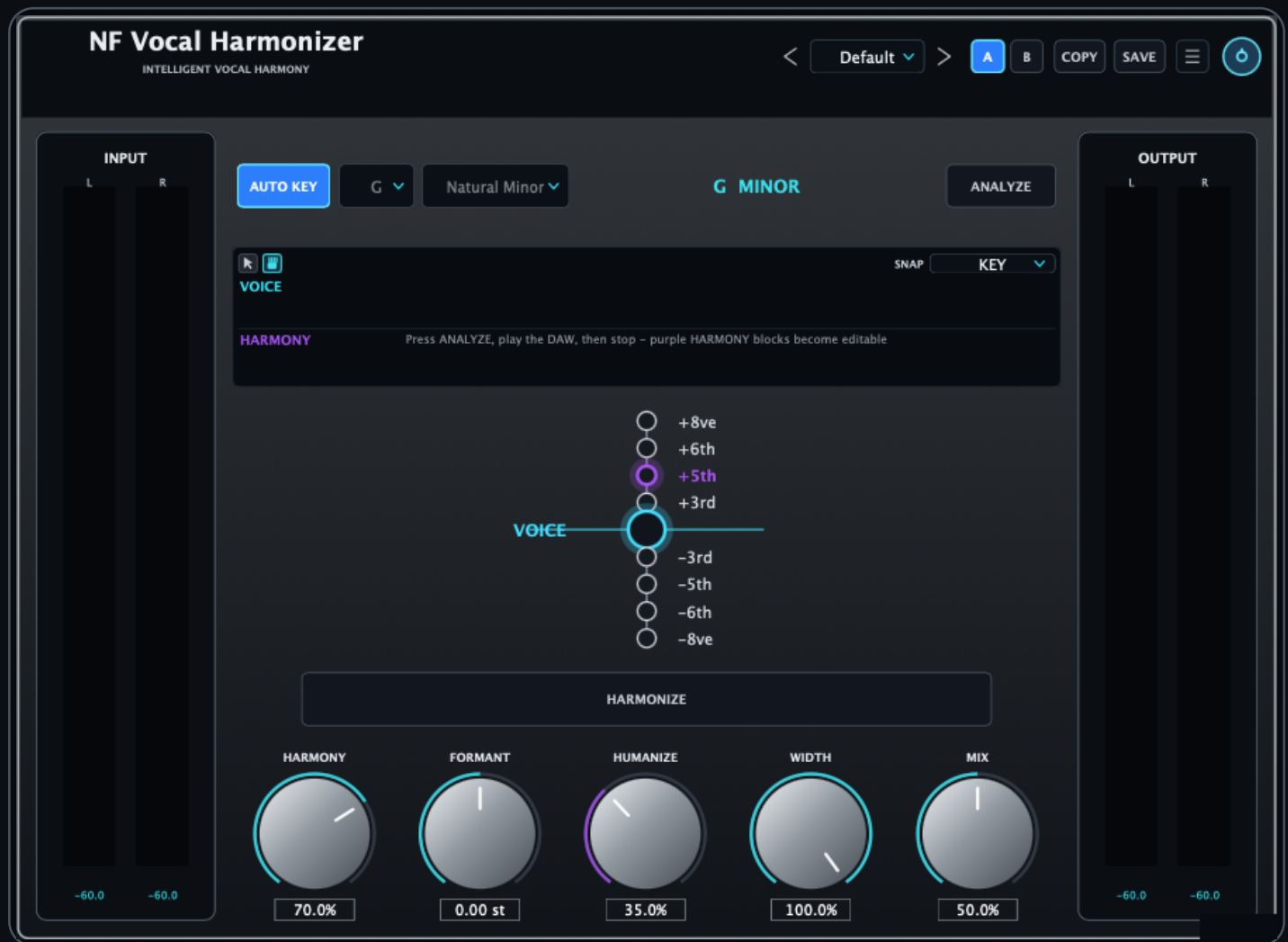


NF VOCAL HARMONIZER

COMPLETE USER MANUAL

Intelligent vocal harmony, surgical editing and immediate musical control.



NF AUDIO TOOLS

BY NENNO FERNANDO

NF AUDIO TOOLS

Contents

A visual guide to producing, analysing and correcting harmonies.

1

Quick start

From a dry vocal to a finished harmony in a few steps.

2

Interface

Every area, button, meter and indicator.

3

Analysis and key

AUTO KEY, Key, Scale and ANALYZE.

4

Note editor

Selection, adjustment, deletion, zoom and navigation.

5

Intervals and controls

Musical choice and finishing the generated voice.

6

Applications

Recipes, professional workflow and troubleshooting.

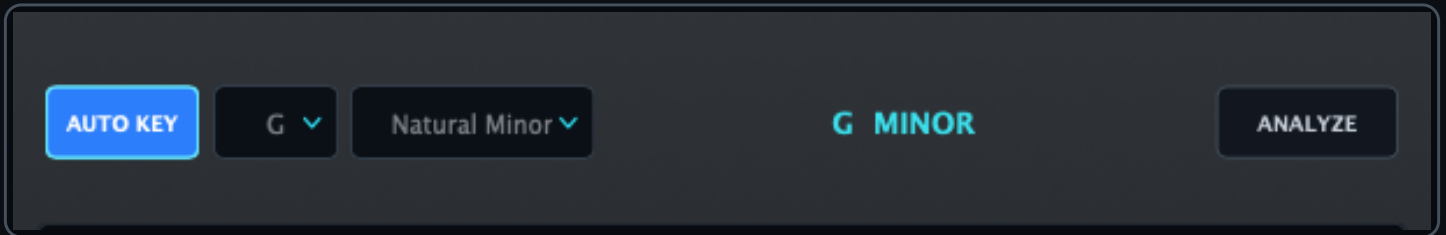
CORE IDEA

The plug-in creates a diatonic harmony voice from a monophonic vocal source. The original voice remains preserved while the harmony can be selected, blended and corrected note by note.

01

Quick start

The recommended path to your first harmony.



1 INSERT

Place the plug-in on a clean, monophonic vocal track.

2 SET THE KEY

Use AUTO KEY or choose Key and Scale manually.

3 ANALYSE

Press ANALYZE, play the passage and stop the DAW when finished.

4 CHOOSE AN INTERVAL

Select a third, fifth, sixth or octave above or below.

5 HARMONIZE

Enable the engine and set HARMONY, FORMANT, HUMANIZE, WIDTH and MIX.

6 CORRECT

Use the editor to adjust or remove only the notes that need intervention.

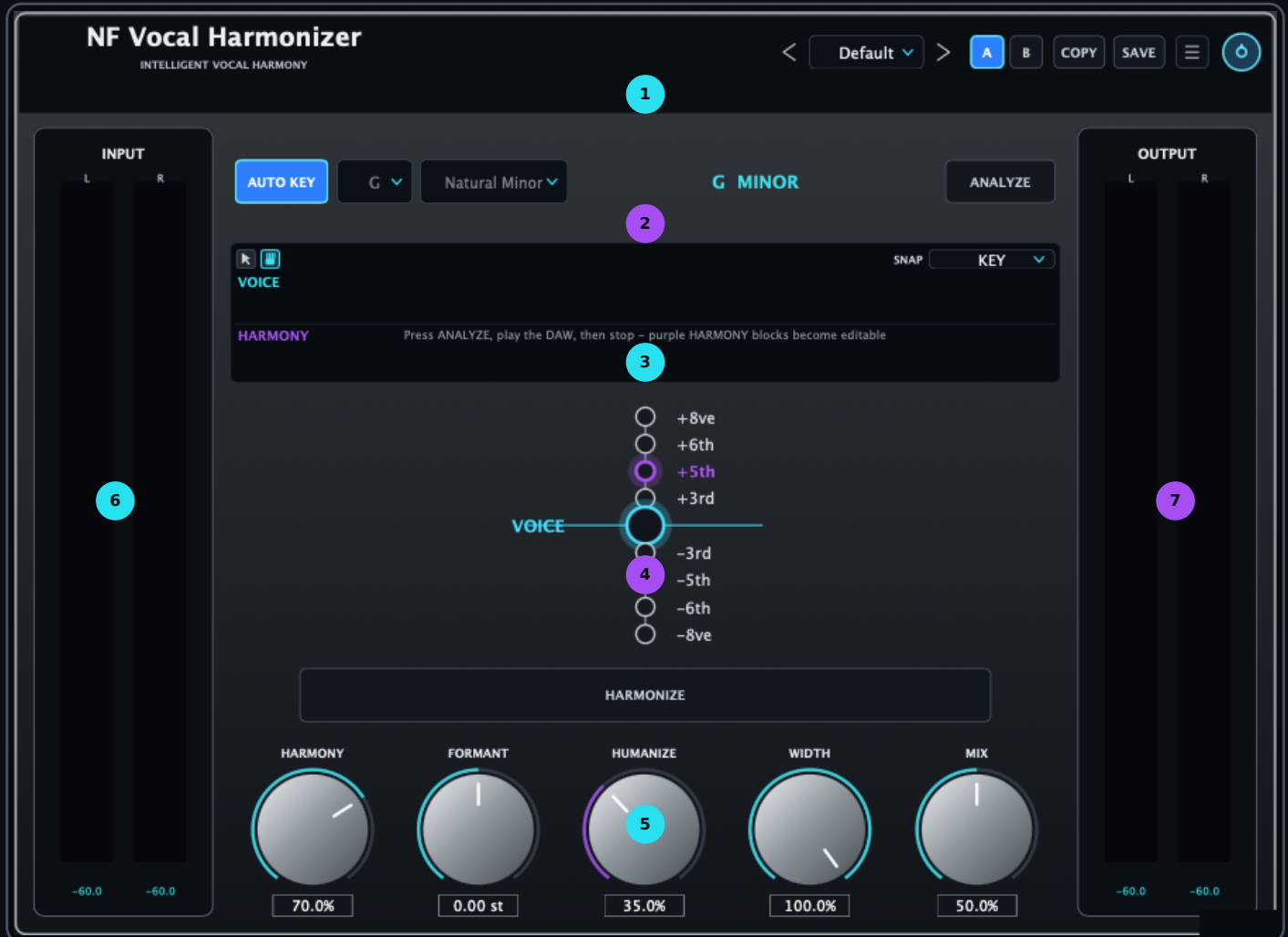
BEST INPUT

An isolated vocal, without simultaneous doubles, with controlled noise and healthy level. Detection is most reliable when one clear fundamental note is present.

02

Interface map

The interface follows the flow: input, musical decision, editing and output.



1 Header and presets

2 Key and ANALYZE

3 VOICE/HARMONY editor

4 Interval rail

5 Finishing controls

6 Input meter

7 Output meter

02.1

Header and presets

State control, comparison and quick access.



< DEFAULT >

Browses, displays and loads presets. The centred name identifies the current preset.

A / B

Keeps two states for immediate comparison.

COPY

Copies the selected state to the other slot.

SAVE

Saves a user preset. Recording-specific edits belong to the session, not to a global preset.

MENU

Opens manuals, the preset folder and product information.

POWER

Enables processing or places it in smooth bypass.

PRESETS vs SESSION

Sound parameters may be stored in presets. The analysed map, deleted notes and surgical corrections should be recalled by the DAW session.

03

Key and analysis

Correct harmony begins with a correct musical reading.

AUTO KEY

G ▼

Natural Minor ▼

G MINOR

ANALYZE

AUTO KEY

Analyses pitch classes and suggests a tonic and mode. Confirm the result by ear.

KEY

Manually sets the tonic from C to B. Use it when the song key is known.

SCALE

Major, Natural Minor, Harmonic Minor, Melodic Minor or Chromatic.

DETECTED KEY

Displays the current reading. For ambiguous passages, prefer manual selection.

ANALYZE

Arms and records the passage. Pulsing green means active analysis; solid green means complete.

IMPORTANT

AUTO KEY is an assistant. Modulations, passing tones and very short passages may require manual Key/Scale.

04

VOICE / HARMONY editor

View the original reading and refine only the generated voice.



VOICE

Cyan blocks represent the detected voice. They are a reference and never erase or alter the DAW recording.

HARMONY

Purple blocks represent the generated harmony. They can be selected, moved in pitch or removed from the result.

SELECT

One click selects a note. Shift-click extends selection. Drag from empty space to select several notes.

SNAP

KEY constrains corrections to the scale; CHROMATIC works in semitones; OFF allows free cent adjustment.

THE EDITOR IS NOT AN AUDIO EDITOR

It stores decisions for the harmony voice. The original waveform, file and DAW region remain untouched.

04.1

Surgical editing

Work on one note or a group without destroying the original voice.



- Click a note to select only that event.
- Drag from empty space to create a rectangular marquee selection.
- Delete/Backspace removes only the selected harmony in that interval.
- The original voice keeps playing and remains visually preserved.
- Command/Ctrl+Z undoes; Shift-Command-Z or Ctrl+Y redoes.
- Short fades prevent clicks at the edges of a removed note.

PITCH CORRECTION

With the selection tool, drag a HARMONY note vertically. Normal movement works in semitones; Shift enables fine adjustment in cents. Double-click restores the automatic decision for that note.

04.2

Zoom and Hand / Pan

Zoom in without losing access to the rest of the performance.



SCROLL

Scroll up to zoom in; scroll down to zoom out.

OPTION / ALT + SCROLL

Moves horizontally through the timeline.

HAND / PAN

Enable the hand, then click-drag left or right.

SELECT

Return to Select before editing or deleting notes.

WHAT MOVES?

Only the viewport moves. Notes are not stretched, shifted in time or changed in the DSP. Zoom changes the visual scale; Pan changes the visible passage.

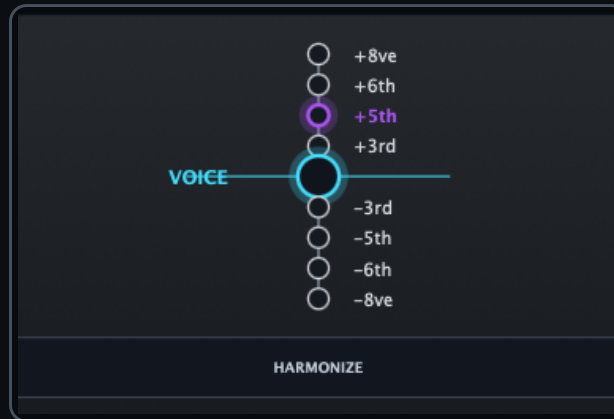
05

Interval rail

One simple choice, applied diatonically within the scale.

ABOVE

- +3rd** Third above: classic harmony double.
- +5th** Fifth above: open and firm.
- +6th** Sixth above: broader colour.
- +8ve** Octave above: brightness and effect.



BELOW

- 3rd** Third below: body and support.
- 5th** Fifth below: depth.
- 6th** Sixth below: darker contrast.
- 8ve** Octave below: low-register effect.

DIATONIC

A +3rd does not always mean four semitones: the engine chooses the appropriate third for each scale degree, following the musical harmony instead of applying a fixed chromatic transpose.

06

Level, tone and naturalness

The first three controls define vocal presence and behaviour.



HARMONY | 0-100%

Controls the generated voice level before the final blend. Use less for invisible support and more for an obvious double.

FORMANT | -12 to +12 st

Compensates vocal colour after transposition. Negative values darken; positive values brighten. Small adjustments generally sound more natural.

HUMANIZE | 0-100%

Adds controlled organic variation between voices. Too much may sound unstable; a small amount reduces mechanical precision.

TIP

When correcting a note, set pitch first; then refine FORMANT and HUMANIZE in the context of the phrase.

06.1

Width, Mix and meters

Finish the stereo image without losing the lead vocal centre.



WIDTH | 0-100%

Defines separation between the left/right harmony variations. Zero moves the harmony toward the centre; 100% creates the widest available image.

MIX | 0-100%

Sets the final balance between the original voice and harmonised result. Always compare in context and at matched perceived level.

INPUT / OUTPUT

Stereo meters confirm activity and help identify excessive level. They do not replace the DAW master meter.

- For lead vocal, begin with moderate WIDTH and low MIX.
- For backing vocals, raise WIDTH and HARMONY before increasing MIX.
- Always check in mono for cancellation.



07

Starting points

Use these as listening references, never fixed rules.

NATURAL THIRD

+3rd | Harmony 55 | Humanize 25 | Width 55 | Mix 25 Subtle support for choruses and sustained phrases.

WIDE BACKING

+3rd/+5th | Harmony 75 | Humanize 45 | Width 100 | Mix 45 More distance from the lead and a wide lateral image.

MODERN OCTAVE

+8ve | Harmony 60 | Humanize 20 | Width 70 | Mix 30 Bright effect; adjust Formant moderately.

LOW SUPPORT

-3rd/-8ve | Harmony 40 | Humanize 20 | Width 35 | Mix 20 Adds body without stealing focus from the lead.

TIGHT DOUBLE

Chromatic | Harmony 35 | Humanize 12 | Width 20 | Mix 18 Near-centre texture for subtle reinforcement.

07.1

Work in stages

Musical decisions first; finishing second.

1**PREPARE THE SOURCE**

Edit noise and confirm there is only one main melody.

2**CONFIRM THE SCALE**

Compare AUTO KEY with the actual song key.

3**CHOOSE THE ROLE**

Decide whether the harmony is support, an obvious backing or an effect.

4**ANALYSE AND EDIT**

Correct only problematic events; preserve natural movement.

5**FINISH IN CONTEXT**

Set level, formant, humanize, width and mix while hearing the full production.

6**SAVE THE SESSION**

Confirm that the map and edits return after closing and reopening the DAW.

08

Shortcuts and troubleshooting

The most important answers during a session.

Click

Selects one note.

Shift-click

Adds/removes from selection.

Drag on empty area

Marquee selection.

Vertical drag

Changes HARMONY pitch.

Shift-drag

Fine cent adjustment.

Delete/Backspace

Removes selected harmony.

Scroll

Horizontal zoom.

Option/Alt-scroll

Horizontal pan.

Hand/Pan drag

Moves the visible window.

Cmd/Ctrl-Z

Undoes the last edit.

TROUBLESHOOTING

No notes in editor

Play a passage with ANALYZE active and audible vocal input.

Wrong key

Disable AUTO KEY and select Key/Scale manually.

Robotic sound

Reduce extreme Humanize/Formant and confirm monophonic detection.

No harmony sound

Check HARMONIZE, HARMONY, MIX and deleted notes.

Plug-in not listed

Rescan and confirm the format installed for the DAW.

09

Installation, compatibility and credits

Essential information for keeping the system reliable.

macOS

Audio Unit, VST3 and AAX. Install the format used by the DAW, then run a fresh scan.

Windows

VST3 and AAX. Audio Unit is available on macOS only.

AAX

Requires a commercially signed build to load in commercial Pro Tools versions.

SESSIONS

Save the session after analysis and verify recall before removing the original recording.

TECHNICAL NOTE

NF Vocal Harmonizer is designed for a monophonic source. Final quality depends on the voice, interval, scale, transposition amount and musical context. Use your ears as the final crit rion.

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